



3AY

CANDIDATE COPY

2026

Roll No.

Time : 3 hours

Maximum Marks : 150

समय : 3 घंटे

अधिकतम अंक : 150

QUESTION PAPER SPECIFIC INSTRUCTIONS

प्रश्न पत्र के लिए विशिष्ट अनुदेश

(Please read the following instructions carefully before attempting questions)

(कृपया प्रश्नों का उत्तर देने से पहले निम्नलिखित निर्देशों को ध्यान से पढ़ें)

1. There are fifteen (15) questions in all.
कुल पंद्रह (15) प्रश्न हैं।
2. All questions are compulsory.
सभी प्रश्न अनिवार्य हैं।
3. No extra/additional sheet will be provided.
कोई अतिरिक्त / विशेष शीट प्रदान नहीं की जाएगी।
4. Answer must be written in the authorized medium. No marks will be given for answers written in a medium other than the authorized one.
उत्तर अधिकृत माध्यम में ही लिखना होगा। अधिकृत माध्यम के अलावा किसी अन्य माध्यम में लिखे गए उत्तरों के लिए कोई अंक नहीं दिए जाएंगे।
5. All questions carry equal marks i.e. 10 each.
सभी प्रश्न समान अंक वाले हैं, अर्थात् प्रत्येक 10 अंक का है।
6. Each Question should be answered within the space provided for that particular question. Otherwise, the answer given will not be evaluated.
प्रत्येक प्रश्न का उत्तर उसके लिए निर्धारित स्थान पर ही लिखा जाए। अन्यथा, दिए गए उत्तर का मूल्यांकन नहीं किया जाएगा।

1. A CPU has a 32 KB direct mapped cache with 128-byte block size. Suppose A is a two-dimensional array of size 512×512 with elements that occupy 8 bytes each. Consider the following two C code segments, P1 and P2:

P1:

```
for (i = 0; i < 512; i++)
{
    for (j = 0; j < 512; j++)
    {
        x += A[i][j];
    }
}
```

P2:

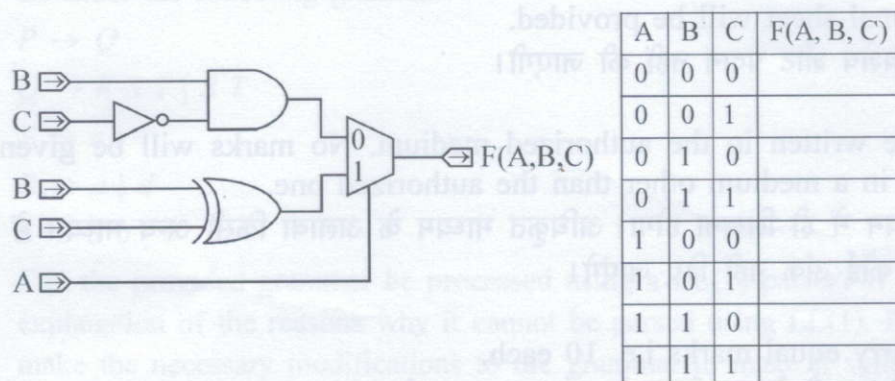
```
for (i = 0; i < 512; i++)
{
    for (j = 0; j < 512; j++)
    {
        x += A[j][i];
    }
}
```

P1 and P2 are executed independently with the same initial state, namely, the array A is not in the cache and i, j, x are in registers. Let the number of cache misses experienced by P1 be M_1 and that for P2 be M_2 . Find

- (a) M_1
 (b) Ratio of M_2/M_1 .

10 Marks

2. An intern at Intel has designed the combinational circuit shown below. His supervisor is unable to comprehend its functionality and has sought your assistance in this matter.



- (a) Fill in the truth table for $F(A, B, C)$ above and express $F(A, B, C)$ in minimal sum-of-products form.
 (b) Is the function $F(A, B, C)$ universal? If so, construct the corresponding circuit with precise input combinations and determine the resulting output.

10 Marks

3. Answer the questions based on the code below with proper justifications.

```
import pickle

import csv

records = [
    {'id': 1, 'name': 'Alice', 'score': 95.5},
    {'id': 2, 'name': 'Bob', 'score': 87.0},
    {'id': 3, 'name': 'Charlie', 'score': 91.3}
]

with open('data.pkl', 'wb') as f:
    pickle.dump(records, f)

with open('data.csv', 'w', newline='') as f:
    writer = csv.DictWriter(f,
        fieldnames=['id', 'name', 'score'])
    writer.writeheader()
    writer.writerows(records)
```

- (i) What is the difference between a text file and a binary file at the byte level? Why can't you read data.pkl in a text editor meaningfully?
- (ii) What is pickling? Explain the security risk of unpickling data from an unknown source. Under what conditions could a malicious pickle file execute arbitrary code?
- (iii) What is the purpose of newline="" when opening a CSV on Windows? What specific problem does omitting it cause?
- (iv) Write a program that reads data.pkl and data.csv, merges by id, adds a 'grade' field (A if score ≥ 90 else B), and writes to 'results.csv'. Include proper exception handling.

10 Marks

4. A python programmer has written the following code involving lists, tuples and dictionaries. Go through it carefully and answer the following questions with proper justifications.

```

a = [1, 2, 3]
b = a
c = a[:]
d = a.copy()
a.append(4)
b[0] = 99
c += [5]
d.insert(1, 0)
t = (1, [2, 3], 4)
t[1].append(5)
print(t)

info = {'name': 'Alice', 'marks': [90, 85, 78]}
info2 = info.copy()
info2['name'] = 'Bob'
info2['marks'].append(95)
print(info)

```

- (i) After all operations, print the values of a, b, c, d. Justify each using Python's memory model.
- (ii) Tuples are immutable. Explain why `t[1].append(5)` does not raise `TypeError`. What is the final value of `t`?
- (iii) `info.copy()` is called yet modifying `info2`'s marks also modifies `info`'s. Explain precisely. How would you fix it?

10 Marks

5. Study the following code fragment and answer every sub-part exhaustively.

```
a = '5'
b = 3.7
c = True
d = None
print(type(int(a) + int(b) + int(c)))
print(a * int(b))
print(int(a) == b)
print(bool(d), bool(0), bool(''), bool(' '), bool([]))
```

- Predict the output of each print statement. For errors, write the error type and state the reason.
- Using only built-in functions, write two different ways to round *b* to the nearest integer and verify that each gives a value 4.
- `int(a) == b` evaluates without error. Explain why Python does not raise `TypeError` here.
- A student claims `int(a)+int(b)+int(c)` and `int(a+b+c)` yield the same result. Prove whether this is true/false with relevant example.

10 Marks

6. A programmer wrote this code to count vowels and consonants. It contains subtle bugs.

```
text = input()
vowels = consonants = 0
i = 0
while i <= len(text):
    ch = text[i]
    if ch.lower in 'aeiou':
        vowels += 1
    elif ch.isalpha():
        consonants -= 1
        i += 2
print(f'Vowels: {vowels}, Consonants: {consonants}')
```

- Identify every bug. For each, state the line, nature of error (syntax/runtime/logical) and its effect.
- Write a corrected version of the program that counts vowels, consonants, digits and spaces using a single while loop.
- Rewrite the program using a “for loop” and “dictionary”. Demonstrate that both versions give same output for ‘Hello World 2026’.
- Explain short-circuit evaluation and show how it can simplify the conditional to a single compound condition without `elif`.

10 Marks

7. A malicious program is designed to replicate itself across a system while evading detection by rewriting its own code on every copy, such that each copy produces a different sequence of machine instructions but computes the exact same result.
- A malicious program hides by injecting its code into the memory space of a legitimate process already running on the operating system. Explain precisely how a process's virtual memory is structured, and at which point during execution this injection becomes possible without the operating system raising an alert.
 - The malicious program rewrites its own instruction sequence on every copy while preserving identical behaviour. Using the concept of semantic equivalence in computation, explain how two syntactically different instruction sequences can produce the same output, and why comparing files by their binary contents alone is fundamentally insufficient as a detection strategy.
 - The program ensures it runs automatically every time the machine restarts, without the user's knowledge. Describe the general mechanism by which an operating system determines which programs to launch at startup, and how a program with sufficient privileges can silently insert itself into this mechanism.
 - A fraudulent email is constructed so that the receiving mail server believes it was sent from a legitimate, trusted address. Explain the three-layer technical verification system used by modern mail servers to authenticate a sender's identity, and identify precisely where a well-crafted fraudulent email can exploit a gap in this verification chain.

10 Marks

8. A message of size 1,200,000 bits is sent from source to destination through a packet-switched network. The message is divided into packets of 12,000 bits each. There are 4 links, each with transmission rate 3 Mbps. Ignore propagation and processing delays. Find the total time to deliver the complete message, assuming store-and-forward switching.

10 Marks

9. A lab has 10 computers. Each computer must be connected to a central network in one of two ways:

Option 1: Bus topology

A single backbone cable of 45 m is used, and each computer requires a 2 m drop line to the backbone.

Option 2: Star topology

Each computer is connected directly to a central switch using 8 m cable.

Find the total cable length required in each topology and state which uses less cable.

10 Marks

10. A web browser requests a webpage from a web server. The page consists of:

- 1 HTML file of size 30 KB
- 6 images, each of size 120 KB

The network throughput is 8 Mbps. Ignore TCP connection setup and latency. Find the total time to download the complete webpage.

10 Marks

11. Table: Attendance(StudentID, SubjectID, Date, Status) where Status is 'P' or 'A'. Write a single query (no window functions) to display: StudentID, total classes, total present, total absent, attendance % rounded to 2 decimal places, and remark 'Detained' if $< 75\%$ else 'Eligible'.

- (i) For a student with 47P out of 63 classes, what is the remark? Show calculation.
- (ii) A student uses `COUNT(Status='P')` to count present days — explain precisely why this is WRONG and what it actually counts.

10 Marks

12. A FinTech startup has been contracted to build a UPI-based digital payment gateway for a nationalized bank. The following constraints and conditions exist:

- The bank's compliance team has not yet finalized the regulatory and security requirements — they expect 4–5 major requirement changes over the next 6 months
- The bank wants a live demonstrable prototype within 45 days to present to RBI auditors
- The development team consists of only 6 developers, all freshers with no prior banking domain experience
- The system must handle 10 lakh transactions/day with 99.99% uptime — making it a high-risk, mission-critical system
- Budget is fixed at ₹80 lakhs with absolutely no room for overrun
- The bank's IT head insists: “We want to see something working every 3 weeks”
- If the system fails post-deployment, the bank faces regulatory penalties and reputational damage

Answer the following:

- (a) Analyze why each of the following models would be unsuitable for this project — give one specific technical reason for each: Waterfall Model, Spiral Model, Pure Agile (Scrum)
- (b) Which single SDLC model or hybrid combination would you recommend? Justify your answer by mapping each project constraint listed above to a specific feature of your recommended model.
- (c) The team lead suggests using RAD (Rapid Application Development) since the prototype is needed in 45 days. Identify two specific reasons why RAD would be dangerous in this context despite the tight deadline, referencing the project's characteristics.
- (d) If the development team switches to your recommended model — what should be the first three activities they perform before writing a single line of code, and why is each critical for this specific project?

10 Marks

13. Suppose that the following processes arrive for execution at the times indicated. Each process will run for the amount of time listed in the given table. Use non-preemptive scheduling and answer the following:

- What is the average turnaround time for these processes with the FCFS scheduling algorithm?
- What is the average turnaround time for these processes with the SJF scheduling algorithm?

Process	Arrival time	Burst Time
P ₁	0.0	8
P ₂	0.4	4
P ₃	1.0	1

10 Marks

14. Let L1 and L2 be two languages over alphabet $\Sigma = \{x, y\}$ as given below:

L1 = Set of all binary strings whose last two symbols are same.

L2 = $\{s_1 x s_2 \mid s_1, s_2 \in \{x, y\}^*, |s_1| = 2, |s_2| \geq 3\}$

Construct a minimum state deterministic finite automaton (DFA) for both L1 and L2. Also write the regular expression.

10 Marks

15. Consider the following grammar:

$P \rightarrow Q$

$Q \rightarrow R S T \mid S T$

$T \rightarrow c \mid \epsilon$

$R \rightarrow a \mid d$

$S \rightarrow S b \mid \epsilon$

Can the provided grammar be processed using a LL(1) parser? If not, provide a detailed explanation of the reasons why it cannot be parsed using LL(1). Based on your analysis, make the necessary modifications to the grammar to make it suitable for LL(1) parsing. Additionally, draw the parse table for the modified grammar and construct a parse tree for the string *dbbbc*.

10 Marks